CS182: Introduction to Machine Learning, Fall 2025
Homework 2

(Due on Friday, Nov. 14 at 11:59 pm (CST))

Notice:

- Please submit your assignments via Gradescope. The entry code is 428YBY.
- Please make sure you select your answer to the corresponding question when submitting your assignments.
- Each person has a total of five days to be late without penalty for all the assignments. Each late delivery less than one day will be counted as one day.

1. [20 points] [*Parameter Estimation*] A 2-axis force sensor, mounted on a robot fingertip, measures an applied force $\mathbf{F} = (F_x, F_y)$. Due to manufacturing tolerances and mechanical crosstalk, the sensor's voltage output V_x is linearly dependent on both F_x and F_y.

   We use a Multivariate Linear Regression Model to calibrate the V_x channel. This model includes a bias term b and two gain terms, m_{11} and m_{12}:

   $$V_x = m_{11}F_x + m_{12}F_y + b$$

   We assume the sensor's measurement noise ϵ is Gaussian, $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Therefore, for the i-th measurement, the observation model is:

   $$V_x^i = (m_{11}F_x^i + m_{12}F_y^i + b) + \epsilon_i$$

   The parameter vector we want to estimate is $\mathbf{w} = \begin{bmatrix} m_{11} \\ m_{12} \\ b \end{bmatrix}$.

   (a) Given a set of N i.i.d. calibration data points $\{X\} = \{(F_x^i, F_y^i), V_x^i\}_{i=1}^N$. Write the Log-Likelihood Function $\mathcal{L}(\mathbf{w}|\{X\})$ for the parameters $\mathbf{w}$. [4 points]

   (b) We know that maximizing $\mathcal{L}(\mathbf{w}|\{X\})$ is equivalent to minimizing the Sum of Squared Errors (SSE) $E(\mathbf{w}) = \sum_{i=1}^N (V_x^i - (m_{11}F_x^i + m_{12}F_y^i + b))^2$.[8 points]

   - Write the SSE $E(\mathbf{w})$ in matrix-vector form: $E(\mathbf{w}) = \|\mathbf{r} - \mathbf{X}\mathbf{w}\|^2$. Clearly define the design matrix $\mathbf{X}$, the parameter vector $\mathbf{w}$, and the observation vector $\mathbf{r}$ based on the model.
   - By computing the gradient of $E(\mathbf{w})$ with respect to the vector $\mathbf{w}$ (i.e., $\nabla_{\mathbf{w}} E(\mathbf{w})$) and setting it to zero, derive the Normal Equations: $(\mathbf{X}^T\mathbf{X})\hat{\mathbf{w}} = \mathbf{X}^T\mathbf{r}$,

   (c) During an experiment, $N = 4$ known forces were applied, and the corresponding V_x voltages were measured.[8 points]

Measurement i	Applied Force (F_x, F_y) (N)	Measured Voltage V_x (V)
1	(1, 0)	2.1
2	(0, 1)	0.9
3	(1, 1)	3.0
4	(2, 0)	4.1

   - Using the data provided in the table, first compute the numerical values for $\mathbf{X}^T\mathbf{X}$ and $\mathbf{X}^T\mathbf{r}$. Then, solve the system to find the optimal parameter estimates $\hat{\mathbf{w}} = (\hat{m}_{11}, \hat{m}_{12}, \hat{b})^T$.
   - A unique least-squares solution $\hat{\mathbf{w}}$ only exists if the applied calibration forces satisfy a specific condition. From an experimental design perspective, what **physical requirement** must the set of $N = 4$ applied forces $\{(F_x^i, F_y^i)\}$ meet to ensure that the parameters (m_{11}, m_{12}, b) can be uniquely determined? Briefly explain why.

Solution:

(a) The PDF for a single measurement i, given the model $V_x^i = (m_{11}F_x^i + m_{12}F_y^i + b) + \epsilon_i$, is:

$$p(V_x^i | F_x^i, F_y^i, \mathbf{w}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(V_x^i - (m_{11}F_x^i + m_{12}F_y^i + b))^2}{2\sigma^2}\right)$$

Assuming N i.i.d. measurements, the total log-likelihood $\mathcal{L}$ is the sum of the individual log-PDFs:

$$\mathcal{L}(\mathbf{w} | \{X\}) = \sum_{i=1}^{N} \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{2\sigma^2}(V_x^i - (m_{11}F_x^i + m_{12}F_y^i + b))^2\right]$$

Ignoring the constant terms (those independent of $\mathbf{w}$), we are left with:

$$\mathcal{L}(\mathbf{w} | \{X\}) \propto -\frac{1}{2\sigma^2} \sum_{i=1}^{N} (V_x^i - (m_{11}F_x^i + m_{12}F_y^i + b))^2$$

(b) First, we define the SSE in matrix-vector form $E(\mathbf{w}) = \|\mathbf{r} - \mathbf{X}\mathbf{w}\|^2$. The components are defined as:

$$\mathbf{X} = \begin{bmatrix} F_x^1 & F_y^1 & 1 \\ F_x^2 & F_y^2 & 1 \\ \vdots & \vdots & \vdots \\ F_x^N & F_y^N & 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} m_{11} \\ m_{12} \\ b \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} V_x^1 \\ V_x^2 \\ \vdots \\ V_x^N \end{bmatrix}$$

To minimize $E(\mathbf{w})$, we find the gradient with respect to $\mathbf{w}$ and set it to zero. We expand $E(\mathbf{w})$:

$$E(\mathbf{w}) = (\mathbf{r} - \mathbf{X}\mathbf{w})^T(\mathbf{r} - \mathbf{X}\mathbf{w})$$

$$E(\mathbf{w}) = (\mathbf{r}^T - \mathbf{w}^T\mathbf{X}^T)(\mathbf{r} - \mathbf{X}\mathbf{w})$$

$$E(\mathbf{w}) = \mathbf{r}^T\mathbf{r} - \mathbf{r}^T\mathbf{X}\mathbf{w} - \mathbf{w}^T\mathbf{X}^T\mathbf{r} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$$

Since $\mathbf{w}^T\mathbf{X}^T\mathbf{r}$ is a scalar, it is equal to its transpose $\mathbf{r}^T\mathbf{X}\mathbf{w}$.

$$E(\mathbf{w}) = \mathbf{r}^T\mathbf{r} - 2\mathbf{r}^T\mathbf{X}\mathbf{w} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$$

Now, we take the gradient with respect to $\mathbf{w}$ (using $\nabla_{\mathbf{x}}(\mathbf{a}^T\mathbf{x}) = \mathbf{a}$ and $\nabla_{\mathbf{x}}(\mathbf{x}^T\mathbf{A}\mathbf{x}) = 2\mathbf{A}\mathbf{x}$ for symmetric $\mathbf{A}$):

$$\nabla_{\mathbf{w}} E(\mathbf{w}) = \nabla_{\mathbf{w}}(\mathbf{r}^T\mathbf{r}) - \nabla_{\mathbf{w}}(2\mathbf{r}^T\mathbf{X}\mathbf{w}) + \nabla_{\mathbf{w}}(\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w})$$

$$\nabla_{\mathbf{w}} E(\mathbf{w}) = \mathbf{0} - 2(\mathbf{r}^T\mathbf{X})^T + 2(\mathbf{X}^T\mathbf{X})\mathbf{w}$$

$$\nabla_{\mathbf{w}} E(\mathbf{w}) = -2\mathbf{X}^T\mathbf{r} + 2(\mathbf{X}^T\mathbf{X})\mathbf{w}$$

Set the gradient to zero to find the minimum $\hat{\mathbf{w}}$:

$$-2\mathbf{X}^T\mathbf{r} + 2(\mathbf{X}^T\mathbf{X})\hat{\mathbf{w}} = 0$$

$$2(\mathbf{X}^T\mathbf{X})\hat{\mathbf{w}} = 2\mathbf{X}^T\mathbf{r}$$

$$(\mathbf{X}^T\mathbf{X})\hat{\mathbf{w}} = \mathbf{X}^T\mathbf{r}$$

(c) First, we construct $\mathbf{X}$ and $\mathbf{r}$ from the data.

$$\mathbf{X} = \begin{bmatrix} F_x^1 & F_y^1 & 1 \\ F_x^2 & F_y^2 & 1 \\ F_x^3 & F_y^3 & 1 \\ F_x^4 & F_y^4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} V_x^1 \\ V_x^2 \\ V_x^3 \\ V_x^4 \end{bmatrix} = \begin{bmatrix} 2.1 \\ 0.9 \\ 3.0 \\ 4.1 \end{bmatrix}$$

Next, we compute the components of the Normal Equations:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 1 & 4 \\ 1 & 2 & 2 \\ 4 & 2 & 4 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{r} = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2.1 \\ 0.9 \\ 3.0 \\ 4.1 \end{bmatrix} = \begin{bmatrix} (2.1 + 0 + 3.0 + 8.2) \\ (0 + 0.9 + 3.0 + 0) \\ (2.1 + 0.9 + 3.0 + 4.1) \end{bmatrix} = \begin{bmatrix} 13.3 \\ 3.9 \\ 10.1 \end{bmatrix}$$

Finally, we solve the 3×3 system $\begin{bmatrix} 6 & 1 & 4 \\ 1 & 2 & 2 \\ 4 & 2 & 4 \end{bmatrix} \begin{bmatrix} m_{11} \\ m_{12} \\ b \end{bmatrix} = \begin{bmatrix} 13.3 \\ 3.9 \\ 10.1 \end{bmatrix}$. (Solving the system of linear equations yields:)

$$\hat{m}_{11} = 2.05$$

$$\hat{m}_{12} = 0.9$$

$$\hat{b} = 0.025$$

Final estimates:

$$\hat{\mathbf{w}} = \begin{bmatrix} \hat{m}_{11} \\ \hat{m}_{12} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 2.05 \\ 0.9 \\ 0.025 \end{bmatrix}$$

Physical Requirement: The applied force points (F_x^i, F_y^i) **must not all lie on the same straight line** in the F_x-F_y plane. We are trying to estimate three parameters (m_{11}, m_{12}, b) which define a 3D plane.

- Mathematically, this requires the design matrix $\mathbf{X}$ to have full column rank (rank 3), which ensures $\mathbf{X}^T\mathbf{X}$ is invertible.
- If all test points (F_x^i, F_y^i) lie on a single line (e.g., $F_y = kF_x + c$), the columns of $\mathbf{X}$ become linearly dependent. This is the problem of multicollinearity.

2. [20 points] [*Parameter Estimation*] Assume that the prior distribution of θ follows a Gaussian distribution:

$$p(\theta) = \mathcal{N}(0, \sigma^2),$$

where θ is an unknown parameter and σ is known. The conditional distribution of x given θ is

$$p(x \mid \theta) = \mathcal{N}(\theta, 1).$$

Please provide detailed derivations for the following:

(a) Derive the marginal distribution $p(x)$. (Hint: $\int_{-\infty}^{\infty} e^{-ax^2+bx+c}dx = \sqrt{\frac{\pi}{a}}e^{\frac{b^2}{4a}+c}$.) [6 points]

(b) Compute the posterior expectation $\mathbb{E}(\theta \mid x)$. [6 points]

(c) Suppose the prior $p(\theta)$ is unknown, and we only know that $p(x \mid \theta) = \mathcal{N}(\theta, \sigma^2)$. Find the posterior expectation $\mathbb{E}(\theta \mid x)$. [8 points]

Solution:

(a)

$$\begin{aligned}
p(x) &= \int_{-\infty}^{\infty} p(x \mid \theta)p(\theta)d\theta \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\theta^2}{2\sigma^2}} d\theta \\
&= \frac{1}{2\pi\sigma} \int_{-\infty}^{\infty} e^{-\frac{\sigma^2(x-\theta)^2+\theta^2}{2\sigma^2}} d\theta \\
&= \frac{1}{2\pi\sigma} \int_{-\infty}^{\infty} e^{-\frac{(\sigma^2+1)\theta^2-2\sigma^2x\theta+\sigma^2x^2}{2\sigma^2}} d\theta \\
&= \frac{1}{2\pi\sigma} \sqrt{\frac{2\pi\sigma^2}{\sigma^2+1}} e^{-\frac{x^2}{2(\sigma^2+1)}} \\
&= \frac{1}{\sqrt{2\pi(\sigma^2+1)}} e^{-\frac{x^2}{2(\sigma^2+1)}} \\
&= \mathcal{N}(0, \sigma^2+1)
\end{aligned}$$

(b) Posterior distribution

$$\begin{aligned}
p(\theta \mid x) &= \frac{p(x \mid \theta)p(\theta)}{p(x)} \\
&= \frac{\mathcal{N}(\theta, 1)\mathcal{N}(0, \sigma^2)}{\mathcal{N}(0, \sigma^2+1)} \\
&= \frac{\frac{1}{2\pi\sigma} e^{-\frac{\sigma^2(x-\theta)^2+\theta^2}{2\sigma^2}}}{\frac{1}{\sqrt{2\pi(\sigma^2+1)}} e^{-\frac{x^2}{2(\sigma^2+1)}}} \\
&= \frac{1}{\sqrt{2\pi\frac{\sigma^2}{\sigma^2+1}}} e^{-\frac{(\theta-\frac{\sigma^2x}{\sigma^2+1})^2}{2\frac{\sigma^2}{\sigma^2+1}}} \\
&= \mathcal{N}\left(\frac{\sigma^2x}{\sigma^2+1}, \frac{\sigma^2}{\sigma^2+1}\right)
\end{aligned}$$

So we have

$$\mathbb{E}[\theta \mid x] = \frac{\sigma^2x}{\sigma^2+1}$$

(c)

$$
\begin{aligned}
\mathbb{E}[\theta \mid x] &= \int_{-\infty}^{\infty} \theta p(\theta \mid x) d\theta \\
&= \int_{-\infty}^{\infty} \theta \frac{p(x \mid \theta) p(\theta)}{p(x)} d\theta \\
&= \frac{\int_{-\infty}^{\infty} \theta p(x \mid \theta) p(\theta) d\theta}{p(x)} \\
&= \frac{\int_{-\infty}^{\infty} \theta \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta) d\theta}{p(x)} \\
&= \frac{\int_{-\infty}^{\infty} \left[\sigma^2 \frac{\theta - x}{\sigma^2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta) + x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta) \right] d\theta}{p(x)} \\
&= \frac{\sigma^2 \int_{-\infty}^{\infty} \frac{\theta - x}{\sigma^2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta) d\theta + \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta) d\theta}{p(x)} \\
&= \frac{\sigma^2 \int_{-\infty}^{\infty} \frac{d}{dx} \left[\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} \right] p(\theta) d\theta + \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta) d\theta}{p(x)} \\
&= \frac{\sigma^2 \int_{-\infty}^{\infty} \frac{dp(x \mid \theta)}{dx} p(\theta) d\theta + \int_{-\infty}^{\infty} x p(x \mid \theta) p(\theta) d\theta}{p(x)} \\
&= \frac{\sigma^2 \frac{d}{dx} \int_{-\infty}^{\infty} p(x \mid \theta) p(\theta) d\theta + x \int_{-\infty}^{\infty} p(x \mid \theta) p(\theta) d\theta}{p(x)} \\
&= \frac{\sigma^2 \frac{dp(x)}{dx} + x p(x)}{p(x)} \\
&= x + \sigma^2 \frac{d}{dx} \log p(x)
\end{aligned}
$$

3. [20 points] [*Parameter Estimation*]

In class, we have learned that the MMSE is achieved when using the conditional expectation $\mathrm{E}[X|\mathbf{Y}]$, i.e., $\arg\min_{\hat{X}} \mathrm{E}[|X-\hat{X}(\mathbf{Y})|^2] = E[X|\mathbf{Y}]$, where X is some unknown random variable and $\mathbf{Y}$ is the observation. However, $\mathrm{E}[X|\mathbf{Y}]$ can be intractable to compute. Now, let's focus on linear estimation and minimize MSE over all linear estimators.

Definition 1 (LLSE)**.**

$$\mathbb{L}[X|\mathbf{Y}] = \arg \min_{linear\ \hat{X}} \mathrm{E}[|X-\hat{X}(\mathbf{Y})|^2]$$

where linear estimators $\hat{X}(\mathbf{Y})$ are of the form

$$\hat{X}(\mathbf{Y}) = a + \sum_{i=1}^{n} b_i Y_i$$

for $a, b_i \in \mathbb{R}$ where $\mathbf{Y} = (Y_1, \cdots, Y_n)$ is a vector of observations.

(a) Define

$$\Sigma_{\mathbf{XY}} = \mathrm{E}[(X-\mu_x)(\mathbf{Y}-\mu_{\mathbf{Y}})^T] \qquad \text{(cross-covariance matrix)}$$
$$\Sigma_{\mathbf{Y}} = \mathrm{E}[(\mathbf{Y}-\mu_{\mathbf{Y}})(\mathbf{Y}-\mu_{\mathbf{Y}})^T] \qquad \text{(covariance matrix)}$$

where $\mu_X = \mathrm{E}(X), \mu_Y = \mathrm{E}[Y] = [\mathrm{E}[Y_1], \cdots, \mathrm{E}[Y_n]]$
Show that

$$\mathbb{L}[X|\mathbf{Y}] = \mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}(\mathbf{Y}-\mu_{\mathbf{Y}}) \tag{1}$$

[5 points]

(b) Show that a linear estimator $\hat{X}(\mathbf{Y})$ achieves minimum mean square error(over all linear estimators) if and only if

$$\mathrm{E}[\hat{X}(\mathbf{Y}) - X] = 0 \tag{2}$$
$$\mathrm{E}[(\hat{X}(\mathbf{Y}) - X)Y_i] = 0 \tag{3}$$

This is called the **orthogonality principle** since it uniquely characterizes $\mathbb{L}[X|\mathbf{Y}]$.
[10 points]

(c) In the real world, data is observed sequentially. How do we efficiently update our LLS estimate on the arrival of new observations? For motivation, start with a simple setting. w.l.o.g., assume $\mathrm{E}[X] = 0$, Suppose observations $Y_1, Y_2, \cdots$ are orthogonal and uncorrelated, i.e., $\langle Y_i, Y_j \rangle = \mathrm{E}[Y_i Y_j] = \mathrm{Cov}(Y_i, Y_j) = 0$ for $i \neq j$ to each other.
For convenience, define

$$\mathbf{Y}^n = (Y_1, \cdots, Y_n)$$
$$\mu_i = \mathrm{E}[Y_i]$$

Show that

$$\mathbb{L}[X|\mathbf{Y}^{n+1}] = \mathbb{L}[X|\mathbf{Y}^n] + \mathbb{L}[X|Y_{n+1}] \tag{4}$$
$$= \mathbb{L}[X|\mathbf{Y}^n] + \frac{\mathrm{Cov}(X, Y_{n+1})}{\mathrm{Var}(Y_{n+1})}(Y_{n+1} - \mu_{n+1}) \tag{5}$$

[5 points]
(Hint: Just check the orthogonality principle!)

Solution:

(a) *Proof.* Let

$$J(a, b_1, \cdots, b_n) = \mathrm{E}\left[\left|X - \left(a + \sum_{i=1}^{n} b_i Y_i\right)\right|^2\right]$$
$$= \mathrm{E}[X^2] - 2a\mathrm{E}[X] - 2\sum_{i=1}^{n} b_i \mathrm{E}[XY_i] + a^2 + 2a\sum_{i=1}^{n} b_i \mathrm{E}[Y_i] + \sum_{i=1}^{n} b_i^2 \mathrm{E}[Y_i^2] + \sum_{i\neq j} b_i b_j \mathrm{E}[Y_i Y_j]$$

Then

$$\frac{\partial J}{\partial a} = 0 \Rightarrow a = \mathrm{E}[X] - \sum_{i=1}^{n} b_i \mathrm{E}[Y_i]$$
$$\frac{\partial J}{\partial b_i} = 0 \Rightarrow \mathrm{E}[XY_i] = a\mathrm{E}[Y_i] + b_i \mathrm{E}[Y_i^2] + \sum_{j\neq i} b_j \mathrm{E}[Y_i Y_j]$$

This is a linear system for a, b_i. To make things easy, assume $\mathrm{E}[X] = \mathrm{E}[Y_i] = 0$ for all $i = 1, \cdots, n$ (You can add the means back later.)
Then

$$a = 0$$
$$\mathrm{E}[XY_i] = \sum_{j=1}^{n} b_j \mathrm{E}[Y_i Y_j],\ i \in \{1, \cdots, n\}$$

Thus,

$$\Sigma_{\mathbf{XY}} = \mathbf{b^T}\Sigma_{\mathbf{Y}} \Rightarrow \mathbf{b^T} = \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}$$

Therefore, when $\mathrm{E}[X] = \mathrm{E}[Y_i] = 0$, we have $\mathbb{L}[X|\mathbf{Y}] = \mathbf{b^T Y} = \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}\mathbf{Y}$
If they are not zero mean, just add the means back in to get

$$\mathbb{L}[X|\mathbf{Y}] = \mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}(\mathbf{Y} - \mu_{\mathbf{Y}})$$

□

(b) *Proof.* $\Rightarrow$ Suppose $\hat{X}(\mathbf{Y}) = \mathbb{L}[X|\mathbf{Y}] = \mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}(\mathbf{Y} - \mu_{\mathbf{Y}})$
Then

$$\begin{aligned}
\mathrm{E}[\hat{X}(\mathbf{Y}) - X] &= \mathrm{E}[\mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}(\mathbf{Y} - \mu_{\mathbf{Y}}) - X] \\
&= \mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}\mathrm{E}[\mathbf{Y} - \mu_{\mathbf{Y}}] - \mathrm{E}[X] \\
&= 0 \\
\mathrm{E}[(\hat{X}(\mathbf{Y}) - X)\mathbf{Y}^T] &= \mathrm{E}[(\mu_X + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}(\mathbf{Y} - \mu_{\mathbf{Y}}) - X)\mathbf{Y}^T] \\
&= \mu_X\mathrm{E}[\mathbf{Y}^T] + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}\mathrm{E}[(\mathbf{Y} - \mu_{\mathbf{Y}})\mathbf{Y}^T] - \mathrm{E}[X\mathbf{Y}^T] \\
&= \mu_X\mathrm{E}[\mathbf{Y}^T] + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}\mathrm{E}[(\mathbf{Y} - \mu_{\mathbf{Y}})(\mathbf{Y} - \mu_{\mathbf{Y}})^T] - \mathrm{E}[X\mathbf{Y}^T] \\
&= \mu_X\mathrm{E}[\mathbf{Y}^T] + \Sigma_{\mathbf{XY}}\Sigma_{\mathbf{Y}}^{-1}\Sigma_{\mathbf{Y}} - \mathrm{E}[X\mathbf{Y}^T] \\
&= -\mathrm{E}[(X - \mu_X)\mathbf{Y}^T] + \mathrm{E}[(X - \mu_X)(\mathbf{Y} - \mu_{\mathbf{Y}})^T] \\
&= -\mathrm{E}[(X - \mu_X)]\mu_{\mathbf{Y}}^T \\
&= 0
\end{aligned}$$

$\Leftarrow$ Suppose

$$\mathrm{E}[\hat{X}(\mathbf{Y}) - X] = 0$$
$$\mathrm{E}[(\hat{X}(\mathbf{Y}) - X)Y_i] = 0$$

Let $\mathcal{U}$ be the set of all linear estimators, for any $u = a + \sum_{i=1}^{n} b_i Y_i \in \mathcal{U}$, we have

$$\mathrm{E}[(\hat{X}(Y) - X)u] = a\mathrm{E}[\hat{X}(Y) - X] + \sum_{i=1}^{n} b_i \mathrm{E}[(\hat{X}(Y) - X)Y_i] = 0$$

Next, we show that $\hat{X}(Y)$ minimize MSE over all linear estimators
For any $v \in \mathcal{U}$

$$\begin{aligned}\mathrm{E}[(X-v)^2] &= \mathrm{E}[(X-\hat{X}(Y))^2] + \mathrm{E}[(\hat{X}-v)^2] + 2\mathrm{E}[(X-\hat{X}(Y))(\hat{X}(Y)-v)] \\ &= \mathrm{E}[(X-\hat{X}(Y))^2] + \mathrm{E}[(\hat{X}-v)^2] \\ &\geq \mathrm{E}[(X-\hat{X}(Y))^2]\end{aligned}$$

Therefore

$$\hat{X}(Y) = \arg \min_{\text{linear } \hat{X}} \mathrm{E}[|X-\hat{X}|^2] = \mathbb{L}[X|\mathbf{Y}]$$

□

(c) *Proof.* Check the orthogonality principle!
It is trivial that $\mathrm{E}[\mathbb{L}[X|\mathbf{Y}^{n+1}] - X] = \mathrm{E}[\mathbb{L}[X|\mathbf{Y}^n]] + \mathrm{E}[\mathbb{L}[X|Y_{n+1}]] = 0$
We just need to show

$$\mathrm{E}[(\mathbb{L}[X|\mathbf{Y}^{n+1}] - X)Y_k] = 0, \ k = 1, \cdots, n+1$$

we know that

$$\mathrm{E}[(\mathbb{L}[X|\mathbf{Y}^{n+1}] - X)Y_k] = \mathrm{E}[(\mathbb{L}[X|\mathbf{Y}^n] + \mathbb{L}[X|Y_{n+1}] - X)Y_k]$$

for $k = n+1$

$$\mathrm{E}[(\mathbb{L}[X|\mathbf{Y}^n] + \mathbb{L}[X|Y_{n+1}] - X)Y_k] = \mathrm{E}[(\mathbb{L}[X|Y_{n+1}] - X)Y_{n+1}] + \mathrm{E}[\mathbb{L}[X|\mathbf{Y}^n]Y_k] = 0$$

This follows from the orthogonality principle and the fact that $\mathbb{L}[X|\mathbf{Y}^n]$ is a linear combination of $Y_1, \cdots, Y_n$ and hence must be orthogonal to Y_{n+1} too.
for $k = 1, \cdots, n$

$$\mathrm{E}[(\mathbb{L}[X|\mathbf{Y}^n] + \mathbb{L}[X|Y_{n+1}] - X)Y_k] = \mathrm{E}[(\mathbb{L}[X|\mathbf{Y}^n] - X)Y_k] + \mathrm{E}[\mathbb{L}[X|Y_{n+1}]Y_k] = 0$$

□

4. [20 points] [*Linear Discriminant Analysis*] We have a two-class dataset with label $\mathcal{Y} = \{-1, +1\}$ and the feature vector $\boldsymbol{x} \in \mathbb{R}^2$. We find the means of these two classes are $\hat{\boldsymbol{\mu}}_{-1} = (-1, -1)^\top$ and $\hat{\boldsymbol{\mu}}_{+1} = (1, 1)^\top$, respectively. And the estimated prior probabilities for these two class are equal (i.e., $\hat{\pi}_{+1} = \hat{\pi}_{-1} = \frac{1}{2}$). Some additional explanations on question 4. The Fisher's linear discriminant is a supervised dimensionality reduction method. The terms Fisher's linear discriminant and LDA are often interchangeably. The Fisher's linear discriminant maximizes the ratio between $\boldsymbol{S}_B$ and $\boldsymbol{S}_W$, where $\boldsymbol{S}_B$ denotes the between-class scatter matrix and $\boldsymbol{S}_W$ denotes the within-class scatter matrix, and this method is based on the so-called Fisher's linear discriminant rule. However, there is another type of LDA, which is based on Bayes discriminant rule, is presented to you in this question. By assuming $p(\boldsymbol{x}|C_i) \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$ (i.e., homoscedasticity), this type of LDA assigns $\boldsymbol{x}$ to the group that maximizes $\pi_i p(\boldsymbol{x}|C_i)$, where π_i represents the prior probability of each classification C_i, and $p(\boldsymbol{x}|C_i)$ represent the i-th density. In question 4(a), we want you to use this type of LDA based on Bayes discriminant rule. In question 4(b), without homoscedasticity, the problem converts to quadratic discriminant analysis (QDA). You can get more detailed information through: `https://harvard-iacs.github.io/2018-CS109A/a-sections/a-section-6/presentation/a_section6.pdf`

   (a) Assuming the covariance martix can be estimated as

$$\hat{\boldsymbol{\Sigma}} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

   with $0 \leq \rho < 1$ when applying LDA. Find the decision boundary as a function of ρ. [7 points]

   (b) Now suppose that we model each class with its own covariance matrix. The covariance matrix of class $+1$ is

$$\hat{\boldsymbol{\Sigma}}_{+1} = \begin{pmatrix} 1/5 & 0 \\ 0 & 5 \end{pmatrix}.$$

   And the covariance matrix of class -1 is

$$\hat{\boldsymbol{\Sigma}}_{-1} = \begin{pmatrix} 5 & 0 \\ 0 & 1/5 \end{pmatrix}.$$

   Describe the decision rule. [8 points]

   (c) When $p(\boldsymbol{x}|C_i) \sim \mathcal{N}\left(\boldsymbol{\mu}_i, \hat{\boldsymbol{\Sigma}}\right)$ (i.e., homoscedasticity), using Fisher's linear discriminant, prove that the vector $\mathbf{w}$ on which the data $\boldsymbol{x}$ are projected is $\mathbf{w} = \hat{\boldsymbol{\Sigma}}^{-1}(\boldsymbol{\mu}_{+1} - \boldsymbol{\mu}_{-1})$. [5 points]

Solution:

   (a) In LDA, decision boundary is $a + \boldsymbol{b}^\top \boldsymbol{x}$, where

$$a = a_{+1} - a_{-1} = \log(\hat{\pi}_{+1}) - \frac{1}{2}\hat{\boldsymbol{\mu}}_{+1}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_{+1}^\top - \log(\hat{\pi}_{-1}) + \frac{1}{2}\hat{\boldsymbol{\mu}}_{-1}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_{-1}^\top.$$

   Hence $a = 0$. Further,

$$\mathbf{b} = \mathbf{b}_{+1} - \mathbf{b}_{-1} = \hat{\boldsymbol{\Sigma}}^{-1}(\boldsymbol{\mu}_{+1} - \boldsymbol{\mu}_{-1}) = \hat{\boldsymbol{\Sigma}}^{-1}(2, 2)^\top = 2/(1+\rho)(1, 1)^\top.$$

   Class $+1$ is chosen over class -1 for $\mathbf{x} = \left(x^{(1)}, x^{(2)}\right)^\top$ iff $a + \mathbf{b}^\top x > 0$, that is iff

$$\frac{2}{1+\rho}\left(x^{(1)} + x^{(2)}\right) > 0.$$

   Equivalently, iff

$$x^{(1)} + x^{(2)} > 0,$$

   which could have been guessed as the solution initially.

   (b) In this case, discriminant function is quadratic, i.e. for $k \in \{-1, +1\}$,

$$\log \hat{\pi}_k \hat{g}(\mathbf{x}) = \text{const} + a_k + \mathbf{b}_k^\top \mathbf{x} + \mathbf{x}^\top \mathbf{c}_k \mathbf{x},$$

   where

$$\mathbf{a}_k = \log \hat{\pi}_k - \frac{1}{2}\log\left|\hat{\boldsymbol{\Sigma}}_k\right| - \frac{1}{2}\hat{\boldsymbol{\mu}}_k^\top \hat{\boldsymbol{\Sigma}}_k^{-1} \boldsymbol{\mu}_k,$$
$$\mathbf{b}_k = \hat{\boldsymbol{\Sigma}}_k^{-1} \hat{\boldsymbol{\mu}}_k,$$
$$\mathbf{c}_k = -\frac{1}{2}\hat{\boldsymbol{\Sigma}}_k^{-1}.$$

Substituting gives

$$
\begin{aligned}
a &= a_{+1} - a_{-1} = \mathbf{0}, \\
\mathbf{b} &= \mathbf{b}_{+1} - \mathbf{b}_{-1} = (5, 1/5)^\top - (-1/5, -5)^\top = \frac{26}{5}(1,1)^\top \\
\mathbf{c} &= \mathbf{c}_{+1} - \mathbf{c}_{-1} = \begin{pmatrix} -12/5 & 0 \\ 0 & 12/5 \end{pmatrix}.
\end{aligned}
$$

Now, class $+1$ is chosen iff $\mathbf{b}_\star^\top \mathbf{x} + \mathbf{x}^\top \mathbf{c}_\star \mathbf{x} > 0$, or equivalently

$$
\begin{aligned}
&\frac{26}{5}\left(x^{(1)} + x^{(2)}\right) - \frac{12}{5}\left(\left(x^{(1)}\right)^2 - \left(x^{(2)}\right)^2\right) > 0 \\
\Leftrightarrow &\left(x^{(1)} + x^{(2)}\right)\left(\frac{13}{6} - x^{(1)} + x^{(2)}\right) > 0.
\end{aligned}
$$

(c) In Fisher's linear discriminant, we need to maximizes the Fisher criterion

$$
\underset{\mathbf{w}}{\text{maximize}} \quad J(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{S}_B \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \mathbf{w}},
$$

which is equivalent to solve

$$
\begin{aligned}
\underset{\mathbf{w}}{\text{maximize}} \quad & \mathbf{w}^\top \mathbf{S}_B \mathbf{w} \\
\text{subject to} \quad & \mathbf{w}^\top \mathbf{S}_W \mathbf{w} = 1
\end{aligned}
$$

Then we can write the Lagrangian function

$$
L(\mathbf{w}, \lambda) = \mathbf{w}^\top \mathbf{S}_B \mathbf{w} + \lambda\left(\mathbf{w}^\top \mathbf{S}_W \mathbf{w} - 1\right).
$$

By setting the derivative of Lagrangian function to zero, we get

$$
\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w}.
$$

We know

$$
\mathbf{S}_B \mathbf{w} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \mathbf{w} = c_1(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2).
$$

Then we get

$$
\mathbf{w} = c\mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2),
$$

where c is some irrelevant constant factor. When $p(\boldsymbol{x}|C_i) \sim \mathcal{N}\left(\boldsymbol{\mu}_i, \hat{\boldsymbol{\Sigma}}\right)$, we know that $\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2 = 2\hat{\boldsymbol{\Sigma}}$, so we have

$$
\mathbf{w} = \frac{c}{2}\hat{\boldsymbol{\Sigma}}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2).
$$

5. [20 points] [*Logistic Discrimination*] Use a quadratic discriminant classifier to classify the K classes $C_1, \ldots, C_K$. Assume that the class-conditional distributions are multivariate normal:

$$p(\mathbf{x} \mid C_i) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$$

where $\boldsymbol{\mu}_i$ is the mean vector and $\boldsymbol{\Sigma}_i$ is the covariance matrix for class C_i.

The quadratic discriminant function is given by:

$$\log \frac{p(\mathbf{x} \mid C_i)}{p(\mathbf{x} \mid C_K)} = \mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0} \tag{6}$$

where $\mathbf{W}_i \in \mathbb{R}^{d \times d}$ is a symmetric matrix related to the covariance matrices, $\mathbf{w}_i \in \mathbb{R}^d$ is a vector, and $w_{i0} \in \mathbb{R}$ is a scalar.

(a) Show the relationship between the parameters $(\mathbf{W}_i, \mathbf{w}_i, w_{i0})$ and the Gaussian parameters $(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$. [5 points]

(b) Using the quadratic discriminants in equation (6), derive the expression for the posterior probability $P(C_i \mid \mathbf{x})$ for each class $i = 1, \ldots, K$. Your answer should involve a softmax function. [5 points]

(c) Show that if all classes share the same covariance matrix, i.e., $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$ for all i, then the decision boundaries become linear. Also, find the expressions for the parameters $\mathbf{W}_i$, $\mathbf{w}_i$, and w_{i0}. [5 points]

(d) Discuss the advantages and disadvantages of quadratic discriminant compared to linear discriminant. Consider aspects such as model flexibility, number of parameters, and risk of overfitting. [5 points]

Solution:

(a) For multivariate normal distributions $p(\mathbf{x} \mid C_i) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, we have:

$$\log p(\mathbf{x} \mid C_i) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{1}{2}\log|\boldsymbol{\Sigma}_i| - \frac{d}{2}\log(2\pi).$$

Expanding the quadratic form:

$$\log p(\mathbf{x} \mid C_i) = -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}_i^{-1}\mathbf{x} + \mathbf{x}^\top \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \frac{1}{2}\boldsymbol{\mu}_i^\top \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \frac{1}{2}\log|\boldsymbol{\Sigma}_i| - \frac{d}{2}\log(2\pi).$$

Now, for the log-ratio:

$$\log \frac{p(\mathbf{x} \mid C_i)}{p(\mathbf{x} \mid C_K)} = \log p(\mathbf{x} \mid C_i) - \log p(\mathbf{x} \mid C_K).$$

Substituting the expressions above, we get:

$$\begin{aligned}\log \frac{p(\mathbf{x} \mid C_i)}{p(\mathbf{x} \mid C_K)} = &-\frac{1}{2}\mathbf{x}^\top \left(\boldsymbol{\Sigma}_i^{-1} - \boldsymbol{\Sigma}_K^{-1}\right)\mathbf{x} + \mathbf{x}^\top \left(\boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \boldsymbol{\Sigma}_K^{-1}\boldsymbol{\mu}_K\right) \\ &- \frac{1}{2}\left(\boldsymbol{\mu}_i^\top \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \boldsymbol{\mu}_K^\top \boldsymbol{\Sigma}_K^{-1}\boldsymbol{\mu}_K\right) - \frac{1}{2}\log\frac{|\boldsymbol{\Sigma}_i|}{|\boldsymbol{\Sigma}_K|}.\end{aligned}$$

Comparing with (6), we identify:

$$\begin{aligned}\mathbf{W}_i &= -\frac{1}{2}\left(\boldsymbol{\Sigma}_i^{-1} - \boldsymbol{\Sigma}_K^{-1}\right) \\ \mathbf{w}_i &= \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \boldsymbol{\Sigma}_K^{-1}\boldsymbol{\mu}_K \\ w_{i0} &= -\frac{1}{2}\left(\boldsymbol{\mu}_i^\top \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \boldsymbol{\mu}_K^\top \boldsymbol{\Sigma}_K^{-1}\boldsymbol{\mu}_K\right) - \frac{1}{2}\log\frac{|\boldsymbol{\Sigma}_i|}{|\boldsymbol{\Sigma}_K|}.\end{aligned}$$

Parameters are learned via MLE:

$$\hat{\boldsymbol{\mu}}_i = \frac{1}{n_i}\sum_{x \in C_i}\mathbf{x}, \quad \hat{\boldsymbol{\Sigma}}_i = \frac{1}{n_i}\sum_{x \in C_i}(\mathbf{x} - \hat{\boldsymbol{\mu}}_i)(\mathbf{x} - \hat{\boldsymbol{\mu}}_i)^\top.$$

(b) According to the Bayes' rule, we have $P(C_i \mid \mathbf{x})\,P(\mathbf{x}) = P(\mathbf{x} \mid C_i)\,P(C_i)$. From the log-odds definition:

$$\log \frac{P(C_i \mid \mathbf{x})}{P(C_K \mid \mathbf{x})} = \mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0} + \log \frac{P(C_i)}{P(C_K)}.$$

Exponentiating both sides:

$$\frac{P\left(C_i \mid \mathbf{x}\right)}{P\left(C_K \mid \mathbf{x}\right)} = \exp\left(\mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0} + \log \frac{P\left(C_i\right)}{P\left(C_K\right)}\right).$$

Let $\eta_i = \mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0} + \log \frac{P(C_i)}{P(C_K)}$, then posterior probability is:

$$P\left(C_i \mid \mathbf{x}\right) = \frac{\exp\left(\eta_i\right)}{1 + \sum_{k=1}^{K-1} \exp\left(\eta_k\right)} \quad \text{for } i = 1, \ldots, K-1.$$

$$P\left(C_K \mid \mathbf{x}\right) = \frac{1}{1 + \sum_{k=1}^{K-1} \exp\left(\eta_k\right)}.$$

This is the softmax function over the quadratic discriminants.

(c) When $\mathbf{\Sigma}_i = \mathbf{\Sigma}$ for all i, the quadratic term $\mathbf{W}_i$ becomes:

$$\mathbf{W}_i = -\frac{1}{2}\left(\mathbf{\Sigma}_i^{-1} - \mathbf{\Sigma}_K^{-1}\right) = \mathbf{0}.$$

Thus, the quadratic term vanishes and the discriminant function becomes linear:

$$\log \frac{p\left(\mathbf{x} \mid C_i\right)}{p\left(\mathbf{x} \mid C_K\right)} = \mathbf{x}^\top \mathbf{\Sigma}^{-1}\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_K\right) - \frac{1}{2}\left(\boldsymbol{\mu}_i + \boldsymbol{\mu}_K\right)^\top \mathbf{\Sigma}^{-1}\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_K\right),$$

which implies the parameters

$$\begin{aligned}
\mathbf{W}_i &= \mathbf{0} \\
\mathbf{w}_i &= \mathbf{\Sigma}^{-1}\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_K\right) \\
w_{i0} &= -\frac{1}{2}\left(\boldsymbol{\mu}_i^\top \mathbf{\Sigma}^{-1} \boldsymbol{\mu}_i - \boldsymbol{\mu}_K^\top \mathbf{\Sigma}^{-1} \boldsymbol{\mu}_K\right).
\end{aligned}$$

(d) **Advantages of quadratic discriminant:**

(1) More flexible decision boundaries (quadratic surfaces)
(2) Better performance when true decision boundaries are nonlinear
(3) Can model class-specific covariance structures

Disadvantages of quadratic discriminant:

(1) Requires more parameters ($O(Kd^2)$ vs $O(d^2)$ for linear discriminant)
(2) Higher risk of overfitting with limited data
(3) More computationally expensive
(4) Less robust to noise in small datasets

linear discriminant is preferred when covariance structures are similar across classes, while quadratic discriminant is better when covariances differ significantly.